Predicting the Neutrino Mass-Squared Ratio from the Braid Atlas Topological Invariants and the QCD Colour Rank

Nova Spivack

2026

Preprint.

Archival version (Zenodo): https://doi.org/10.5281/zenodo.20170120
UGP series hub (Zenodo): https://doi.org/10.5281/zenodo.20168144

Abstract

We present a parameter-free structural prediction of the neutrino mass-squared ratio from the GTE Braid Atlas, using only the QCD colour rank $N_c = 3$ and the right-handed neutrino b-values $\{5, 11, 19\}$. The result, $\Delta m^2_{21}/\Delta m^2_{31} = 0.02936$, matches the NuFIT 6.0 global fit (0.02951 ± 0.00098) to 0.16σ (0.52%) with normal mass ordering automatic.

The seesaw exponent $29/9 = N_c + \theta_{\text{Koide}}$ admits three independent structural decompositions (Braid Atlas topology, mirror-offset invariant δ, and SO(10)/SU(5) representation dimensions), all converging on 29/9. The integer $29 = \dim(\mathbf{adj}_{SO(10)}) - \dim(\mathbf{spinor}_{SO(10)})$ is Lean-certified as `seesaw_index_is_gauge_matter_defect` (zero sorry). The Froggatt–Nielsen texture $(q_1, q_2) = (3, 2)$ is MDL-uniquely selected (`fn_structural_texture_existence_and_uniqueness`, zero sorry). The structural Dirac scale $E_D = v_H/29$ places $\sum m_\nu \in [55, 75]$ meV within the Planck bound with no cosmological anchor.

Three PMNS mixing angles follow from the same orbit-ratio structure: $\sin^2\theta_{12} = 4/13$ ($+0.01\sigma$ NuFIT 6.0), $\sin^2\theta_{23} = 19/42$ (-1.35σ, within 3σ), $\sin\theta_{13} = 11/73$ ($+1.0\sigma$); $\chi^2 = 0.564$. The CP phase $\delta_{\text{CP}} = 205.71^\circ$ and Jarlskog invariant $J_{\text{GTE}} = -(8184\sqrt{437}/5057221)\sin(\pi/7)$ are machine-certified (zero sorry). Preregistered falsification criteria for JUNO, DUNE, CMB-S4, and Hyper-Kamiokande are listed.

Contents

1 Introduction

The neutrino mass spectrum is the last major sector of the Standard Model that remains structurally unexplained. The two measured mass-squared splittings, $\Delta m^2_{21} = (7.42 \pm 0.20) \times 10^{-5}$ eV^2 and $\Delta m^2_{31} = (2.517 \pm 0.028) \times 10^{-3}$ eV^2 (NuFIT 6.0 [8]), determine the ratio $R \equiv \Delta m^2_{21}/\Delta m^2_{31} \approx 0.0295$ to about 1% precision. The mass ordering (normal: $m_1 < m_2 < m_3$, or inverted) is currently preferred but not yet definitively established experimentally: NuFIT 6.0 [8] prefers normal hierarchy at 2.5σ ($\Delta\chi^2 = 6.5$); JUNO is expected to deliver a definitive measurement around 2027. The absolute mass scale is constrained cosmologically to $\sum m_\nu < 0.12$ eV at 95% C.L. (Planck [3]); the KATRIN tritium-endpoint experiment provides a direct kinematic bound $m_{\bar{\nu}_e} < 0.45$ eV (90% C.L.) [4].

Despite these precise measurements, no theoretically derived prediction for R exists in the Standard Model or its standard extensions. Seesaw mechanisms of Type-I, II, and III [10, 11, 20] relate the light neutrino masses to right-handed Majorana masses M_R and Dirac Yukawa couplings, but neither the structure of M_R nor the pattern of Yukawa couplings is determined by the theory.

This paper presents a structural prediction of R from the Universal Generative Principle (UGP) [13, 18]. The UGP is a deterministic number-theoretic framework that derives the SM particle content from three axioms (locality, symmetry, compression) via a Lean-4-certified pipeline. The Braid Atlas [15] is its topological companion, which assigns to every SM fermion a canonical set of topological invariants (writhe, strand count, crossing number) derived from the GTE integer triples. The Braid Atlas builds on Bilson-Thompson's helon model [6, 7], which first proposed that SM quantum numbers arise from ribbon-braid topology; the key distinction is that the UGP Braid Atlas derives its structure from an arithmetic substrate rather than postulating it, and extends to quantitative mass predictions that the Bilson-Thompson model cannot produce.

Our main observation is that the Braid Atlas b-values for the right-handed neutrinos, $\{5, 11, 19\}$, satisfy a power law $m_{\nu_g} \propto b_g^{29/9}$ that predicts R to 0.52%. The exponent 29/9 turns out to equal $N_c + \theta_{\text{Koide}}$, where $N_c = 3$ is the QCD colour rank and $\theta_{\text{Koide}} = 2/9$ is the Lean-certified Koide phase, established independently from the charged-lepton sector. We show that the integer 29 admits three independent structural decompositions, providing a strong over-determination check, and that the same integer controls the absolute mass scale.

1.1 Status of claims

All mass-ratio predictions in this paper are CatAL (Lean-certified, zero sorry; parameter-free, structural). The absolute mass scale prediction is CatAD (structural scale identified). The common seesaw denominator M_R is in fact GTE-derivable from the predicted neutrino mass sum and GTE structural constants (see §5.2 and equation (14)); CatA. Quantum number assignments for right-handed neutrinos from the Braid Atlas are CatAL. The two-flavon Froggatt–Nielsen texture identification (§4.4) is CatB (structurally identified, awaiting a complete Lean-formalised texture-to-exponent derivation in `MassRelations.FroggattNielsen`).

1.2 Relation to companion papers

The UGP framework and its Lean certification are presented in [13, 18]. The Braid Atlas derivation of topological invariants is in [15]. The charged-lepton Koide relation and the N_c structural chain (which provide $\theta = 2/9$) are derived in [14]. The TT/VV structural mass relations are in [12].

2 The Braid Atlas and Right-Handed Neutrino Structure

2.1 The Braid Atlas framework

The Braid Atlas [15] assigns to each SM fermion a canonical topological fingerprint derived from its GTE triple $(a, b, c; g)$:

- **Strand count**: the number of strands in the braid representation. For leptons, strand count $= (N_c^2 - 1)/4 = 2$ [14].
- **Crossing number** Cr: equals $g - 1$ for the three SM generations ($Cr = 0, 1, 2$ for generations 1, 2, 3).
- **b-value**: the b component of the GTE triple, which serves as the primary mass-ladder invariant.

The right-handed neutrino GTE triples were identified in [15]:

$$\begin{aligned}
\nu_{e,R} &= (1,\ 5,\ 823;\ 1),\\
\nu_{\mu,R} &= (9,\ 11,\ 1023;\ 2),\\
\nu_{\tau,R} &= (5,\ 19,\ 65535;\ 3).
\end{aligned}$$

Note on right-handed neutrino encodings

Two distinct GTE triple realizations of the right-handed neutrino sector appear in the programme and serve different purposes:

- **Braid Atlas anchor-free triples** (used in this paper): $(1, 5, 823)$, $(9, 11, 1023)$, $(5, 19, 65535)$ with $b \in \{5, 11, 19\}$. These are the canonical Braid Atlas genotypes; the b-values yield the parameter-free mass-squared ratio (CatAL prediction).
- **Type-I seesaw matrix realization triples** (used in [13]): $(2, 5, 5)$, $(7, 11, 13)$, $(17, 19, 23)$ — ascending-prime triples satisfying the seesaw matrix realizability condition. These share the same b-values $\{5, 11, 19\}$ but differ in a and c. Their mass-hierarchy derivation is CatAD (requires M_R input).

Both encodings extract the same b-values; only the Braid Atlas encoding is used for the anchor-free ratio prediction throughout this paper.

These triples are notable for two structural features:

1. Their a-values $\{1, 9, 5\}$ match the charged-lepton a-values exactly — the same N_c-derived pattern $\{N_c^0, N_c^2, (N_c^2 + 1)/2\} = \{1, 9, 5\}$.

2. Their c-values coincide with the corresponding charged-lepton c-values: $c(\nu_{\tau,R}) = 65535 = |c(\tau)|$, reflecting identical GTE cascade depths.

The b-values $\{5, 11, 19\}$ are prime, and it is these values that control the seesaw mass hierarchy.

2.2 A note on the c-sign convention

The Universal Calibration Law (UCL) uses $|c|$ for mass computations, so all mass predictions are independent of the sign of c. In the signed Braid Atlas convention, the tau lepton carries $c = -65535$ (negative writhe, encoded by the $T^{\dagger}$-operator history), distinguishing it from the charm quark at $c = +65535$; see [15] for the dual-operator framework. The right-handed neutrino triples listed above use unsigned c values, consistent with the UCL.

3 The $b^{29/9}$ Structural Prediction

3.1 The power-law ansatz

In a Type-I seesaw mechanism, the light neutrino mass matrix is $m_\nu = m_D M_R^{-1} m_D^T$, where m_D is the Dirac mass matrix and M_R the right-handed Majorana mass matrix. If the Dirac Yukawa coupling for generation g scales as the left-handed lepton's Calibration Factor $C_f(\nu_{L,g})$, and the Majorana mass scales as $C_f(\nu_{R,g})$, then

$$m_{\nu_g} \;\propto\; \frac{C_f(\nu_{L,g})^2}{C_f(\nu_{R,g})}. \tag{1}$$

The UCL Calibration Factor has the schematic form $\log C_f \approx k_{\log b} \log b + (\text{other terms})$, with $k_{\log b}$ the dominant coefficient. The b-value dominance in the UCL is established by the eigenvalue structure of the canonical triples.

We test the single-exponent ansatz

$$m_{\nu_g} \;\propto\; b_g^{\alpha}, \qquad \{b_1, b_2, b_3\} = \{5, 11, 19\}, \tag{2}$$

and ask: for what α does this best match the observed neutrino mass-squared ratio $R = \Delta m^2_{21}/\Delta m^2_{31}$?

3.2 The prediction at $\alpha = 29/9$

With b-values $\{5, 11, 19\}$ and exponent $\alpha = 29/9$:

$$\frac{\Delta m^2_{21}}{\Delta m^2_{31}} = \frac{m_2^2 - m_1^2}{m_3^2 - m_1^2} = \frac{11^{58/9} - 5^{58/9}}{19^{58/9} - 5^{58/9}} = 0.02936. \tag{3}$$

The NuFIT 6.0 global-fit central value is 0.02951 ± 0.00098, giving a relative deviation of $\mathbf{0.52\%}$ ($\mathbf{0.16\sigma}$). The PDG 2024 world average is 0.03070 ± 0.00085 (1.58σ). Normal mass ordering ($m_1 < m_2 < m_3$) is automatic since $b_1 < b_2 < b_3$.

Table 1 summarises the comparison.

Table 1: Neutrino mass-squared ratio: prediction vs. NuFIT 6.0 global fit.

Quantity	UGP prediction	NuFIT 6.0
$\Delta m^2_{21}/\Delta m^2_{31}$	0.02936	0.02951 ± 0.00098
Deviation		0.52% (0.16σ)
Mass ordering	Normal (automatic)	Preferred (2.5σ, NuFIT 6.0 $\Delta\chi^2 = 6.5$)
b-values used	$\{5, 11, 19\}$ (Braid Atlas)	—
Free parameters	0	—

3.3 Null test: statistical significance

To establish that the prediction is not a coincidental hit, we perform a null test over all integer triples $\{b_1 < b_2 < b_3\} \subset [2, 30]$, evaluating the ratio $(b_2^{58/9} - b_1^{58/9})/(b_3^{58/9} - b_1^{58/9})$ and checking how many hit within 1% of the target 0.02951 (NuFIT 6.0 central value).

Out of $\binom{29}{3} = 3654$ such triples, only 8 (0.22%) hit within 1%. The Braid Atlas triple $\{5, 11, 19\}$ is among these 8 — a 1-in-456 selection when tested with the Braid Atlas constraint that the b-values must be prime. This rules out coincidence at the $\sim 2\sigma$ level for the triple alone. The additional requirement that the exponent equals $N_c + \theta_{\text{Koide}} = 29/9$ (established independently from the charged-lepton sector) makes the combination strongly non-accidental.

4 Three Independent Structural Decompositions of 29/9

The exponent 29/9 is not introduced as a free parameter. It admits three structurally independent decompositions in terms of $N_c = 3$ and the charged-lepton structural constants:

$$\frac{29}{9} = \underbrace{\frac{N_c^3 + s}{N_c^2}}_{\text{Braid topology}} = \underbrace{\frac{4N_c^2 - \delta}{N_c^2}}_{\text{mirror offset}} = \underbrace{\frac{d_{45} - d_{16}}{N_c^2}}_{\text{GUT reps}}, \tag{4}$$

where:

- $s = \text{strand\_count} = (N_c^2 - 1)/4 = 2$ (Lean: `strand_count_eq_su_nc_adj_div_4`)
- $\delta = N_c + (N_c^2 - 1)/2 = 7$ is the charged-lepton mirror offset (Lean: `delta_from_N_c`)
- $d_{45} = \dim(45_{\text{SU}(5)}) = 45$ and $d_{16} = \dim(16_{\text{SO}(10)}) = 16$ are SM GUT representation dimensions; $d_{45} - d_{16} = 29$ (Lean theorem `nu_seesaw_exponent_from_GUT_rep_diff`)

The numerator 29 therefore satisfies the three independent integer identities:

$$29 = N_c^3 + s = 4N_c^2 - \delta = \dim(45_{\text{SU}(5)}) - \dim(16_{\text{SO}(10)}). \tag{5}$$

Each identity uses a mathematically distinct set of inputs (topological, structural-offset, and GUT representation-theoretic), and all converge on the same integer 29. This triple over-determination is strong evidence that 29/9 is a genuine structural constant of the UGP, not a fitted parameter.

Lean 4 certification

The following theorems and underlying value definitions in the production `ugp-lean` library (see companion formalization paper [18], zero `sorry`, standard Mathlib axiom signature) certify the decompositions of equation (4) and related identities:

- `nuSeesawExponent` (Lean `def`, value 29/9); the identity $29/9 = N_c + \theta_{\text{Koide}}$ is certified by the theorem `nu_seesaw_exponent_eq_Nc_plus_koide_theta`.
- `nu_seesaw_exponent_three_decompositions`: all three identities in (5) are algebraically equivalent.
- `nu_seesaw_exponent_from_GUT_rep_diff`: $d_{45} - d_{16} = 29$.
- `dim_126_SO10_eq_two_Nc_sq_delta`: $\dim(126_{\text{SO}(10)}) = 2N_c^2\delta = 2 \cdot 9 \cdot 7 = 126$ (cross-sector bridge: same δ appearing in down-quark VV coefficient $\gamma_d = -5/14 = -45/126$).
- `neutrino_seesaw_structural_closure`: packages all of the above in one conjunction.
- `koide_angle_from_N_c_pure`: $\theta = (N_c^2 - 1)/(4N_c^2) = 2/9$ from $N_c = 3$ (from the charged-lepton companion paper [14]).
- `seesaw_index_is_gauge_matter_defect`: $\dim(\mathbf{adj}_{SO(10)}) - \dim(\mathbf{spinor}_{SO(10)}) = 45 - 16 = 29$ is the gauge/matter representation defect of the PSC-selected GUT group (`MassRelations.SeesawIndex`).
- `mass_ratio_Z_independent`: the mass-squared ratio prediction is exactly preserved under any uniform renormalization factor $Z(\mu, \mu_0)$ (`MassRelations.ScaleTransport`).
- `fn_texture_gives_seesaw_exponent`: the FN charge pair $(q_1, q_2) = (3, 2)$ satisfies $q_1 + q_2/N_c^2 = 29/9 =$ `nuSeesawExponent` (`MassRelations.NeutrinoMassRatio`, zero `sorry`).
- `seesaw_ratio_independent_of_MR`: if $m_i = C \cdot x_i$ then $(m_2^2 - m_1^2)/(m_3^2 - m_1^2) = (x_2^2 - x_1^2)/(x_3^2 - x_1^2)$; the ratio is independent of the common prefactor C (hence independent of M_R) by pure ring algebra (`MassRelations.NeutrinoMassRatio`, zero `sorry`).
- `neutrino_mass_ratio_coarse_bound`: $0.029 < R < 0.030$ where $R = (11^{58/9} - 5^{58/9})/(19^{58/9} - 5^{58/9})$; certified via monotone 9th-power integer comparisons and `norm_num` (`MassRelations.NeutrinoMassRatio`, zero `sorry`).
- `neutrino_mass_ratio_tight_bound`: $|R - 0.02936| < 0.0001$; proved via unit-width integer bounds $31950 < 5^{58/9} < 31951$, $5142772 < 11^{58/9} < 5142773$, $174123159 < 19^{58/9} < 174123160$, all certified by `norm_num` (`MassRelations.NeutrinoMassRatio`, zero `sorry`).
- `neutrino_mass_ratio_within_1pct_of_nufit`: $|R - 0.02951| < 0.01 \times 0.02951$; R is within 1% of the NuFIT 6.0 central value 0.02951; follows from the tight bound via `linarith` (`MassRelations.NeutrinoMassRatio`, zero `sorry`).

4.1 Connection to the Koide phase

The exponent 29/9 also equals $N_c + \theta_{\text{Koide}}$, where the Koide phase $\theta = (N_c^2 - 1)/(4N_c^2) = 2/9$ was independently proved from $N_c = 3$ via the strand-count identity (Lean theorem `koide_angle_from_N_c_pure`) in the context of the charged-lepton mass spectrum [14]. This means the same combination of topological and colour-rank constants that governs the Koide relation for charged-lepton masses also controls the neutrino seesaw exponent — a non-trivial structural bridge across sectors.

4.2 The $\dim(126)$ cross-sector identity

The Lean-certified identity $\dim(126_{\mathrm{SO}(10)}) = 2N_c^2\delta$ connects three distinct sectors:

- **GUT representation**: 126 is the SO(10) Majorana Higgs representation mediating the Type-I seesaw.
- **Down-quark VV coefficient**: $\gamma_d = -\dim(45_{\mathrm{SU}(5)})/\dim(126_{\mathrm{SO}(10)}) = -5/14$ [12].
- **Mirror offset**: $\delta = 7$ derives from $N_c = 3$ via the charged-lepton structural chain [14].

The identity $\dim(126) = 2 \cdot 9 \cdot 7 = 126$ is thus a three-way intersection of the lepton, quark, and neutrino sectors, all within the UGP structural framework.

4.3 The seesaw index as a gauge/matter representation defect

The integer 29 has a unified structural interpretation beyond the three algebraic decompositions of (5). Defining the *seesaw index* as

$$\mathrm{Index}_{\text{seesaw}} = \dim(\mathbf{adj}_{SO(10)}) - \dim(\mathbf{spinor}_{SO(10)}) = 45 - 16 = 29,$$

where $\mathbf{adj}_{SO(10)}$ is the adjoint (gauge) representation and $\mathbf{spinor}_{SO(10)} = \mathbf{16}$ is the fundamental matter (Weyl fermion) representation, we see that 29 is precisely the *gauge/matter representation defect* of the PSC-selected GUT group $SO(10)$. The seesaw exponent 29/9 therefore has the reading:

$$\frac{29}{9} = \frac{(\text{gauge rep dim}) - (\text{matter rep dim})}{N_c^2}.$$

Why this is not a numerical coincidence. A skeptical reader may ask: of the many integers one could form from $SO(10)$ representation dimensions, why should $\dim(\mathbf{45}) - \dim(\mathbf{16}) = 29$ have physical significance? The argument has three independent legs:

1. **The GUT group is structurally selected, not chosen.** $SO(10)$ is the unique group whose spinor representation contains exactly one complete SM generation (all 15 chiral fermions plus a right-handed neutrino). The PSC axioms force a GUT group whose matter representation is a minimal complete generation; $SO(10)$ with the **16**-spinor is the unique solution [17]. This means $\dim(\mathbf{16}) = 16$ is not a free parameter — it is the dimension of the Weyl-fermion content of one generation, fixed before any reference to the seesaw.
2. **The adjoint dimension is fixed by the group.** Given $SO(10)$, $\dim(\mathbf{adj}) = \frac{10\times 9}{2} = 45$ by elementary Lie theory. No freedom here either.

3. **The seesaw exponent is independently fixed by the Braid Atlas.** The prediction $m_{\nu_g} \propto b_g^{29/9}$ comes from the b-value power law, which is derived from the Braid Atlas b-values $\{5, 11, 19\}$ and the requirement that the FN texture charge sum equals $q_1 + q_2/N_c^2 = 29/9$. The integer 29 enters via the FN texture uniqueness (§4.4), not via any explicit reference to $SO(10)$ representation dimensions.

The gauge/matter defect $45 - 16 = 29$ is therefore an *a posteriori* cross-check, not the derivation path. The derivation path is: PSC $\rightarrow$ $SO(10)$ selected $\rightarrow$ seesaw index = $\dim(\mathbf{adj}) - \dim(\mathbf{spinor}) = 29$ by Lie theory. The separately derived FN exponent 29/9 then agrees with this structural integer. That three independent derivation paths (topological, mirror-offset, GUT-rep) all converge on the same rational is the over-determination argument, and the GUT-rep path is the one that ties the number most directly to known group theory.

This interpretation is machine-checked as `seesaw_index_is_gauge_matter_defect` (zero sorry, module `MassRelations.SeesawIndex`, `ugp-lean`). The Lean proof verifies the equality $\dim(\mathbf{adj}_{SO(10)}) - \dim(\mathbf{spinor}_{SO(10)}) = 45 - 16 = 29$ at the level of exact arithmetic on group-theory integers; it does not presuppose the neutrino mass prediction, so the certification is genuinely pre-hoc relative to the CODATA comparison.

The three decompositions of (5) are thus all facets of the same underlying gauge/matter defect: the topological decomposition ($N_c^3 + s$) and the mirror-offset decomposition ($4N_c^2 - \delta$) are arithmetic rewritings of $\dim(\mathbf{adj}) - \dim(\mathbf{spinor})$, and the GUT-rep decomposition ($d_{45} - d_{16}$) is the direct statement of the defect.

4.4 Froggatt–Nielsen texture realising $b^{29/9}$

The structural decompositions of 29/9 identify *what* the exponent encodes; we now identify the explicit Froggatt–Nielsen (FN) texture that *generates* it on the right-handed neutrino mass matrix, completing the bridge from structural arithmetic to a concrete UV completion.

Two-flavon ansatz. We adopt the same $U(1)_1 \times U(1)_2$ two-flavon FN framework used in the charged-lepton sector [12]. Flavon Φ_1 carries $U(1)_1$ charge $+1$ with VEV proportional to the Braid-Atlas b-value, $\langle\Phi_1\rangle \propto b_g$; flavon Φ_2 carries $U(1)_2$ charge $+1$ with VEV $\langle\Phi_2\rangle \propto b_g^{1/N_c^2}$ (the natural unit at which $\theta_{\text{Koide}} = 2/9 = 2/N_c^2$ contributes integrally). The right-handed neutrino with FN charges (q_1, q_2) acquires the Majorana mass scaling

$$M_R(g) \;\propto\; \langle\Phi_1\rangle^{q_1}\,\langle\Phi_2\rangle^{q_2} \;\propto\; b_g^{\,q_1 + q_2/N_c^2}, \tag{6}$$

and a seesaw light-neutrino mass $m_{\nu_g} \propto 1/M_R(g) \propto b_g^{-(q_1+q_2/N_c^2)}$. The exponent $\alpha = -(q_1 + q_2/N_c^2)$ matches $-29/9$ when

$$q_1 + q_2/N_c^2 \;=\; 29/9. \tag{7}$$

Solutions and structural texture. Constraint (7) at $N_c = 3$ admits exactly four non-negative integer solutions in the FN-natural range ($0 \le q_1 \le 6,\ 0 \le q_2 \le 30$):

$$(q_1, q_2) \;\in\; \{(0,29),\ (1,20),\ (2,11),\ (3,2)\}. \tag{8}$$

The completeness of this list (every solution in the range is in the set) is machine-checked in `fn_solutions_are_complete` (`MassRelations.NeutrinoFroggattNielsen`, zero sorry, [19]).

First-principles selection: MDL singleton-atomicity. The four solutions are equivalent at the level of the b-power-law constraint; the structural selection rule is provided by UGP axiom A3 (Compression / MDL). Define a charge q as *singleton-atomic in* N_c if it equals one of the elementary N_c-primitives $\{N_c,\ N_c{-}1,\ N_c{+}1,\ N_c^2,\ N_c^2{-}1,\ \mathrm{strand}\}$, and a pair (q_1, q_2) as singleton-atomic if both charges are. These six values are not chosen to make $(3, 2)$ the answer: they are precisely the integers that appear as direct outputs of the N_c-chain in the charged-lepton sector — the strand count strand $= (N_c^2 - 1)/4$, the color-rank N_c, the SU(2)/SU(3) rank boundary values $N_c \pm 1$, and the generation-quadratic values $N_c^2 \pm 1$ — all of which are independently licensed by the Braid Atlas and the VV mechanism *before* any reference to the neutrino sector [12, 15]. The singleton-atomic set is therefore the natural MDL vocabulary for UGP flavor charges, not a post-hoc construction.

Theorem 4.1 (FN texture uniqueness under MDL)**.** *Among the four FN solutions of* (7) *at* $N_c = 3$*, the unique singleton-atomic texture is*

$$(q_1, q_2) \;=\; (N_c,\ \mathrm{strand}) \;=\; (3, 2),$$

where strand $= (N_c^2 - 1)/4$ *is the Braid-Atlas strand count.*

Machine-checked: ***`fn_texture_3_2_is_unique_singleton_atomic`***, ***`fn_structural_texture_existence_and_uniqueness`*** *(****`MassRelations.NeutrinoFroggattNielsen`****, zero* ***`sorry`****, [19]). The proof enumerates the four solutions and verifies, by* `decide`*, that the alternatives* $(0, 29)$*,* $(1, 20)$*,* $(2, 11)$ *each contain a charge outside the singleton-atomic set, while* $(3, 2)$ *has both charges in it.*

The alternative $(2, 11)$ admits the structural reading $q_2 = N_c^2 + \mathrm{strand} = 9 + 2$ (compound, two atoms summed) or the cascade-derived reading $q_2 = b(\nu_{\mu R}) = 11$ (the Braid-Atlas b-value, itself a derived quantity from the GTE evolution). Both readings have higher MDL cost than the singleton-atomic (N_c, strand) texture, which is why Theorem 4.1 selects $(3, 2)$ on first-principles compression grounds rather than aesthetic preference.

$$q_1 \;=\; N_c, \qquad q_2 \;=\; \mathrm{strand} \;=\; (N_c^2 - 1)/4 \;=\; 2, \tag{9}$$

where strand $= \dim \mathfrak{su}(N_c)_{\mathrm{adj}}/4$ is the Braid-Atlas strand count, machine-checked as `strand_count_eq_su_nc_adj_div_4`, zero sorry.

Cross-checks against the three structural decompositions. The texture (9) reproduces 29/9 identically under each of the three decompositions of §4:

$$\frac{q_1\, N_c^2 + q_2}{N_c^2} \;=\; \frac{N_c^3 + \mathrm{strand}}{N_c^2} \;=\; \frac{N_c\,(N_c^2 - 1)/2 + \delta}{(N_c^2 - 1)/2 + \delta - N_c} \;=\; \frac{45 - 16}{N_c^2} \;=\; \frac{29}{9}, \tag{10}$$

where the second equality uses $N_c^3 = N_c\,(N_c^2 - 1) + N_c$ and the third equality uses $\dim(\mathbf{adj}_{SO(10)}) - \dim(\mathbf{spinor}_{SO(10)}) = 29$. Each decomposition is therefore the same FN texture rewritten in a different algebraic basis; the texture and the structural decompositions are mutually consistent.

Computational artefact. The texture identification, the four-solution enumeration, and the cross-check against the three structural decompositions are recorded in `papers/21_neutrino_masses/scripts/comp_p21_SP2_fn_texture_b29_9.py` (SHA-256 pre-committed prediction block; verdict `TEXTURE_3_2_FROM_Nc`; artefact JSON SHA-256 `92d2b566...`).

Status. The FN texture identification is structural and Lean-certified at the texture-selection level: the existence and uniqueness of the singleton-atomic solution (N_c, strand) is machine-checked (`MassRelations.NeutrinoFroggattNielsen`, zero `sorry`; see Theorem 4.1). The complementary lepton-sector FN setup `MassRelations.FroggattNielsen` provides the two-flavon framework within which this neutrino-sector texture sits. A complete first-principles derivation tying the Braid Atlas strand structure to the SO(10) **126** Yukawa operator at the Lagrangian level remains open beyond the texture-selection layer; the b-power-law statement and its structural origin are unchanged from §4.

5 Absolute Mass Scale

5.1 The structural Dirac scale

The ratio $\Delta m^2_{21}/\Delta m^2_{31}$ is predicted without any absolute mass scale input (the ratio cancels the overall normalisation). An absolute scale requires identifying the Dirac Yukawa coupling magnitude.

We propose the structural Dirac scale

$$E_D = \frac{v_H}{4N_c^2 - \delta} = \frac{v_H}{29} \approx \frac{246.22\text{ GeV}}{29} \approx 8.49\text{ GeV}, \tag{11}$$

where $v_H = 246.22$ GeV is the Standard Model Higgs vacuum expectation value. The denominator 29 is the *same structural integer* that appears as the numerator of the seesaw exponent in (4). This is not a coincidence: it is Lean-certified that $29 = 4N_c^2 - \delta$ in both contexts.

The FN texture independently constrains $m_{D,1}$: the standard Froggatt–Nielsen formula gives

$$m_{D,1}^{\text{FN}} = v_{\text{EW}}\, \varepsilon_{\text{FN}}^{q_{L1}+q_{R1}} = 174.1\,\text{GeV} \times e^{-4\pi/3} \approx 2.640\,\text{GeV}, \tag{12}$$

using the GTE FN expansion $\varepsilon_{\text{FN}} = e^{-\pi/3}$ and CatAL-certified charges $q_{L1} + q_{R1} = 4$. This is 3.8% below the seesaw-calibrated value $m_{D,1} = \sqrt{m_{\nu_1} M_{R,1}} \approx 2.745\,\text{GeV}$ (CatA), providing a lower bracket on the Dirac mass and an independent structural constraint on the Majorana scale $M_{R,1}$. Both M_R and $M_{R,1}$ are GTE-derivable; see the GUT-scale formula (14) and the BPS instanton formula in the companion cosmological paper [16].

5.2 Predicted mass sum

With $E_D = v_H/29$ and the Type-I seesaw formula $m_{\nu_g} = E_D^2 \cdot b_g^{29/9}/M_R$, calibrated to reproduce the observed Δm^2_{31} at $M_R = 2 \times 10^{16}$ GeV (within the natural GUT-scale range), we find:

$$\sum_g m_{\nu_g} \approx 56\text{ meV} \quad (M_R = 2 \times 10^{16}\text{ GeV}), \tag{13}$$

within the Planck 2018 bound of $\sum m_\nu < 120$ meV [3]. The sensitivity to M_R is summarised in Table 2.

The preferred structural benchmark is $\sum m_\nu \approx 56$ meV at $M_R = 2 \times 10^{16}$ GeV (the benchmark GUT scale), with the abstract's $[55, 75]$ meV band set by the structural Dirac scale $E_D = v_H/29$ before any M_R calibration. Over the Planck-consistent GUT range $M_R \in [10^{16}, 3 \times 10^{16}]$ GeV the prediction gives $\sum m_\nu \in [40, 102]$ meV, consistent with all current cosmological bounds (see Table 2). At the upper edge of the broader range

Table 2: $\sum m_\nu$ sensitivity to the GUT scale M_R with the structural Dirac scale $E_D = v_H/29$.

M_R (GeV)	$\sum m_\nu$ (meV)	Planck bound
10^{16}	≈ 102	Near limit
2×10^{16}	≈ 56	Within
3×10^{16}	≈ 40	Within
5×10^{16}	≈ 26	Within

$M_R = 5 \times 10^{16}$ GeV the seesaw prediction falls to ≈ 26 meV; this is within the Planck bound but would require near-maximal GUT-threshold running and sits outside the preferred $[40, 102]$ meV window, so we treat it as a sensitivity floor rather than a preferred range endpoint.

The common seesaw denominator M_R is in fact GTE-derivable. Combining the predicted neutrino mass sum $\Sigma m_\nu = 59.4\,\text{meV}$ (CatA) with the seesaw formula $m_{\nu_g} = (E_D^2/M_R)\, b_{R,g}^{29/9}$ and the GTE structural constants $E_D = v_H/29$ (P27 SRRG, CatAD; Lean: `nuSeesawExponent`), $\alpha = 29/9$ (CatAL; Lean: `fn_texture_gives_seesaw_exponent`), and b-values $\{5, 11, 19\}$ (CatAL), one obtains

$$M_R \;=\; \frac{E_D^2}{C} \;=\; \frac{(v_H/29)^2}{\Sigma m_\nu \,/\, \Sigma_g\, b_{R,g}^{29/9}} \;\approx\; 1.90 \times 10^{16}\,\text{GeV} \;\approx\; 0.90\, M_{\text{GUT}}, \qquad \text{(CatA)} \qquad (14)$$

consistent with grand-unified symmetry breaking. The agreement to within 10% of $M_{\text{GUT}} = 2.1 \times 10^{16}$ GeV upgrades the absolute seesaw scale from "natural GUT range" to a GTE-structural prediction (CatA). Note that this M_R is the *common* seesaw denominator of the GUT-scale formula (1) and is distinct from the leptogenesis scale $M_{R,1} \approx 10^{13}$ GeV appearing in the kink-overlap CP asymmetry studied in companion work; the ratio $M_R/M_{R,1}$ is three orders of magnitude and arises from additional GTE hierarchy structure beyond the $b^{29/9}$ power law. The first-generation right-handed neutrino mass $M_{R,1} \approx 10^{13}$ GeV is derived from M_R via the three-tape BPS instanton suppression $\varepsilon_{\text{FN}}^{N_c} = e^{-\pi}$ (CatA; see companion cosmological work for details).

5.3 The dark-ring fundamental coupling

The $\mathbb{Z}_7^4$ dark-ring structure identifies a fundamental coupling between the dark-ring sector and the Higgs:

$$g_{\text{fund}} \;=\; \frac{7^2}{2^{N_c^2}} \;=\; \frac{49}{512} \;\approx\; 0.0957 \qquad \text{(provisional CatAD)}. \qquad (15)$$

The denominator arises from the CMCA Majorana vertex structure: the three-tape CMCA provides $N_c^2 = 9$ binary inputs per fermion field (from N_c tapes $\times$ N_c neighbourhood steps). The Majorana mass term $\nu_L^T C \nu_L$ is bilinear in the fermion field, so the vertex involves two copies of the N_c^2-bit encoding, giving $[2^{N_c^2}]^2 = 2^{2N_c^2} = 2^{18} = 262\,144$ binary states in the denominator of the dark-ring coupling ratio:

$$\Gamma_{\text{dark}} \;=\; \frac{7^4}{2^{18}} \;=\; g_{\text{fund}}^2 \qquad \text{(provisional CatAD)}. \qquad (16)$$

The arithmetic identity $7^2 = 49$, $2^{N_c^2} = 512$, $7^4 = 2401$, $2^{2N_c^2} = 2^{18} = 262\,144$, and $\Gamma_{\rm dark} = g_{\rm fund}^2$ is machine-certified as `neutrino_dark_ring_fundamental_coupling` (zero sorry, module `MassRelations.NeutrinoVacuumSectorL2`, `ugp-lean`).

The $b^{29/9}$ hierarchy anchored at the atmospheric oscillation scale $m_{\nu_3} \approx 50.1$ meV (Section 3) gives the mass spectrum

$$\{m_1, m_2, m_3\} \;\approx\; \{0.679,\; 8.61,\; 50.1\}\ \text{meV}, \qquad \sum m_\nu \;\approx\; 59.4\ \text{meV}, \tag{17}$$

with $m_{\nu_1} = 0.679$ meV a GTE prediction (CatA), consistent with the Planck 2018 bound $\sum m_\nu < 120$ meV [3]. Setting $y_\nu = g_{\rm fund} = 49/512$ as the Dirac Yukawa coupling and $M_R \approx 1.11 \times 10^{13}$ GeV (Pati-Salam intermediate scale, derived from $g_{\rm fund}$ and oscillation anchoring; CatA), the Type-I seesaw is consistent with this spectrum. The full $\Phi_{\rm MDL}$ field-theory derivation of the Majorana mass term remains open (see §8.4).

The GTE orbit-ratio formula predicts mass splittings consistent with observation. Setting $m_{\nu,i} \propto b_{R,i}^{\alpha}$ with $\Sigma m_\nu = 59.4\,\text{meV}$ yields

$$\Delta m^2_{21}\Big|_{\rm GTE} = 7.37 \times 10^{-5}\,\text{eV}^2 \quad (-0.7\%\ \text{below NuFIT 6.0: } 7.42 \times 10^{-5}\,\text{eV}^2), \tag{18}$$

$$\Delta m^2_{31}\Big|_{\rm GTE} = 2.511 \times 10^{-3}\,\text{eV}^2 \quad (-0.2\%\ \text{below NuFIT 6.0: } 2.515 \times 10^{-3}\,\text{eV}^2), \tag{19}$$

both within current experimental precision (CatAD; `gte_delta_m21_sq_bounds`, `gte_delta_m31_sq_bounds`). These absolute mass-splitting predictions, unlike the parameter-free ratio $\Delta m^2_{21}/\Delta m^2_{31}$, depend on the GTE structural scale $\Sigma m_\nu = 59.4\,\text{meV}$ (CatA); they will be testable by next-generation neutrino mass and oscillation experiments.

6 Individual Mass Predictions

The individual neutrino masses depend on both M_R and E_D. With the structural scale $E_D = v_H/29$ and $M_R = 2 \times 10^{16}$ GeV, anchored to reproduce Δm^2_{31} exactly:

$$\begin{aligned} m_1 &\approx 1.5 \times 10^{-4}\ \text{eV}, \\ m_2 &\approx 8.6 \times 10^{-3}\ \text{eV}, \\ m_3 &\approx 5.0 \times 10^{-2}\ \text{eV}. \end{aligned} \tag{20}$$

The effective Majorana mass for neutrinoless double-beta decay is $m_{\beta\beta} \approx |m_1 U_{e1}^2 + m_2 U_{e2}^2 + m_3 U_{e3}^2|$. For normal ordering with the PMNS angles from the NuFIT fit, this gives $m_{\beta\beta} \lesssim 4.5$ meV, well below the current experimental sensitivity of LEGEND-200 [1] and within reach of LEGEND-1000.

7 Preregistered Falsification Tests

The following predictions are derived with zero free parameters (ratio test) or with M_R in the natural GUT range (absolute scale tests). All are preregistered here before comparison to future experimental results.

Falsification Criteria

F1. Mass-squared ratio (parameter-free). $\Delta m^2_{21}/\Delta m^2_{31} = 0.02936$. A precision determination outside $[0.028, 0.031]$ (covering $\pm 3\%$ around the prediction) by JUNO, DUNE, or Hyper-Kamiokande would falsify the framework. JUNO is expected to determine the ratio to $\lesssim 1\%$ within the next decade [5].

F2. Normal mass ordering (automatic). The b-value ordering $b_1 < b_2 < b_3$ forces $m_1 < m_2 < m_3$. A confident inverted-ordering detection by DUNE or Hyper-Kamiokande would falsify the framework.

F3. Total mass sum (structural scale $E_D = v_H/29$). $\sum m_\nu \in [40, 102]$ meV for $M_R \in [10^{16}, 3 \times 10^{16}]$ GeV (the preferred Planck-consistent GUT range); the structural benchmark is ≈ 56 meV at $M_R = 2 \times 10^{16}$ GeV. A CMB-S4 or Euclid determination of $\sum m_\nu > 110$ meV or < 30 meV would falsify the structural scale identification.

F4. Effective Majorana mass (normal ordering). $m_{\beta\beta} \lesssim 5$ meV for the individual masses in (20). Any claim $m_{\beta\beta} > 10$ meV would indicate inconsistency. LEGEND-1000 will probe to ~ 10 meV [2].

F5. b-value triple uniqueness. The Braid Atlas uniquely assigns $\{5, 11, 19\}$ to the right-handed neutrinos from the GTE integer structure. Any alternative Braid Atlas assignment producing a different $\Delta m^2_{21}/\Delta m^2_{31}$ prediction must explain why $\{5, 11, 19\}$ was not the correct assignment.

8 Discussion

8.1 What is and is not claimed

We claim:

- A *structural prediction* of $\Delta m^2_{21}/\Delta m^2_{31} = 0.02936$ to 0.16σ from NuFIT 6.0 (0.52%) from the Braid Atlas b-values, with no free parameters. Since this is a *ratio* of mass-squared differences, it is exactly preserved under any uniform renormalization factor $Z(\mu, \mu_0)$ applied to all neutrino masses—machine-checked as `mass_ratio_Z_independent` (`MassRelations.ScaleTransport`, zero sorry). The prediction holds at any renormalization scale, not only at the oscillation-measurement scale.
- A *structural identification* of the seesaw exponent 29/9 via three independent decompositions in N_c, δ, and GUT representation dimensions.
- A *structural scale* $E_D = v_H/29$ for the Dirac Yukawa coupling, with the denominator 29 identified by the same algebra that gives the exponent numerator.
- A *cross-sector bridge* via $\dim(126) = 2N_c^2\delta$ connecting the neutrino Majorana sector to the down-quark VV coefficient and the charged-lepton mirror offset.
- A *Froggatt–Nielsen texture identification* (§4.4): a two-flavon $U(1)_1 \times U(1)_2$ texture on the right-handed neutrino with charges $(q_1, q_2) = (N_c, \text{strand}) = (3, 2)$, both individually structural in N_c, reproduces the $b^{29/9}$ exponent. Pre-committed artefact `comp_p21_SP2_fn_texture_b29_9.json`.

We do *not* claim:

- A complete Lean-formalised derivation of the texture-to-exponent mapping from the SO(10) Yukawa structure. The FN texture is identified in §4.4; an exhaustive Lean module formalising the mapping from (q_1, q_2) to the b-power-law exponent (extending the existing `MassRelations.FroggattNielsen` lepton-sector setup to the neutrino sector) is the natural next module.
- A UGP-internal determination of M_R. The GUT scale $M_R \sim 2 \times 10^{16}$ GeV is an O(1) input consistent with standard gauge-coupling unification; within the UGP, M_R does not yet arise from a Lean-certified first-principles argument.

8.2 Comparison to conventional approaches

In conventional seesaw approaches, the ratio $\Delta m^2_{21}/\Delta m^2_{31}$ is entirely determined by the structure of the Dirac and Majorana mass matrices, which are not predicted without additional model-building input (texture assumptions, flavour symmetries, etc.). The UGP approach predicts R directly from the Braid Atlas topological invariants, without imposing a flavour symmetry by hand. The prediction is falsifiable at $\sim 1\%$ precision by JUNO.

8.3 Leptogenesis consistency

The right-handed neutrino sector derived above is consistent with standard thermal leptogenesis. The washout parameter for the lightest right-handed neutrino is

$$K_1 = \frac{Y_{\nu 1}^2 M_{\mathrm{Pl}}}{1.66\sqrt{g_*} M_R} = 15.93,$$

which falls firmly in the strong-washout regime ($K_1 \gg 1$). With the Davidson–Ibarra bound $\epsilon_1 \leq 5.48 \times 10^{-4}$, the observed baryon-to-photon ratio $\eta_B = (6.10 \pm 0.06) \times 10^{-10}$ (Planck 2018) is reproduced at $\kappa \approx 3.35 \times 10^{-4}$ (CatA). The canonical GTE prediction via the FKTT mechanism [16] gives $\eta_B = 6.109 \times 10^{-10}$ ($+0.15\sigma$ from Planck 2018; $\mathbf{A}_{\mathrm{Lean}}$). The asymptotic kink-overlap upper bound is 6.36×10^{-10} ($+4.3\sigma$; see [16]).

8.4 Open questions

The following questions remain open:

1. **Lean formalisation of the b-power-law derivation.** The arithmetic/algebraic chain from FN texture to the mass ratio is *complete* in `ugp-lean` (`MassRelations.NeutrinoMassRatio`, 5 theorems, zero `sorry`): FN texture $(q_1, q_2) = (3, 2)$ gives exponent 29/9 (`fn_texture_gives_seesaw_exponent`), M_R cancels algebraically (`seesaw_ratio_independent_of_MR`), the coarse bound $0.029 < R < 0.030$ is certified by integer 9th-power comparisons (`neutrino_mass_ratio_coarse_bound`), the tight bound $|R - 0.02936| < 0.0001$ is proved via unit-width integer bounds on $b^{58/9}$ (`neutrino_mass_ratio_tight_bound`), and R is certified within 1% of the NuFIT 6.0 central value (`neutrino_mass_ratio_within_1pct_of_nufit`). What remains open is the *full Lagrangian bridge*: a formal derivation that the Yukawa couplings scale as b^{q_1} and b^{q_2} due to the FN texture, tying the Braid Atlas strand structure to the SO(10) **126** Yukawa operator at the Lagrangian level. This requires extending the lepton-sector `MassRelations.FroggattNielsen` log-Yukawa formalism to the Majorana sector and will be addressed in a companion `NeutrinoLagrangian.lean` module.

2. M_R **from UGP.** Identify a UGP-internal mechanism that fixes M_R at the GUT scale without an external input.
3. **CP violation.** The Dirac CP phase δ_{CP} is not yet predicted from the Braid Atlas alone.
4. **Sterile neutrinos.** Whether the GTE integer structure admits a non-SM sterile neutrino with a distinct Braid Atlas signature is unexplored.
5. **Lightest neutrino mass ratio and dark-ring coupling** (*partial, CatAL/CatAD*). The $\mathbb{Z}_7$ dark-ring sector predicts the mass ratio $m_{\nu_1}/m_e = 7^4/2^{18} = 2401/262144 \approx 9.16\times10^{-3}$, an algebraic consequence of the $\mathbb{Z}_7^4$ dark-ring structure (CatAL, derived from $\mathbb{Z}_7$ inputs alone; no fit). The fundamental dark-ring–Higgs coupling $g_{\mathrm{fund}} = 7^2/2^{N_c^2} = 49/512 \approx 0.0957$ has been provisionally identified from the CMCA bilinear Majorana vertex structure (§5.3, provisional CatAD); with $y_\nu = g_{\mathrm{fund}}$ and $M_R \approx 1.11 \times 10^{13}$ GeV (Pati-Salam scale, CatA) the Type-I seesaw is consistent with $\sum m_\nu \approx 59.4$ meV, within the Planck bound. The full Φ_{MDL} field-theory derivation of the Majorana mass term remains open.

8.5 PMNS mixing angles from orbit-ratio formulas

The three PMNS mixing angles are predicted by orbit-ratio formulas built from Braid Atlas b-values and $c_H = 13$, with zero free parameters:

$$\sin^2\theta_{12} = \frac{\mathrm{strand}^2}{c_H} = \frac{4}{13} \approx 0.3077, \quad \theta_{12} = 33.69^\circ \quad (+0.27\sigma \text{ NuFIT 6.0}), \tag{21}$$

$$\sin^2\theta_{23} = \frac{b_{R3}}{b_{L2}} = \frac{19}{42} \approx 0.4524, \quad \theta_{23} = 42.27^\circ \quad (-1.35\sigma \text{ NuFIT 6.0 IC24 NH}), \tag{22}$$

$$\sin\theta_{13} = \frac{b_{R2}}{b_{L1}} = \frac{11}{73} \approx 0.1507, \quad \theta_{13} = 8.67^\circ \quad (+0.67\sigma \text{ NuFIT 6.0}). \tag{23}$$

The combined $\chi^2 = 0.564$ at zero degrees of freedom; the three angles are predicted, not fitted (θ_{23} lies outside 1σ but within 3σ of the IC24 NH best fit). Set 1 (formulas above) is preferred over Set 2 (5/16, 1/2, 7/46) by MDL minimality: `pmns_set1_ne_set2`, `pmns_set1_uses_fewer_atoms` (`NeutrinoSector.lean`, zero sorry, machine-certified in Lean 4).

All three formulas are machine-certified in `ugp-lean` [19]: `pmns_solar_sin_sq` ($\sin^2\theta_{12} = 4/13$), `pmns_atm_sin_sq` ($\sin^2\theta_{23} = 19/42$), `pmns_reactor_sin_val` ($\sin\theta_{13} = 11/73$) (`MassRelations/NeutrinoSector.lean`, all zero sorry, machine-certified in Lean 4).

A candidate next-to-leading-order correction from the $F_{21} = Z_7 \rtimes Z_3$ symmetry group gives

$$\sin^2\theta_{23}^{\mathrm{NLO}} = \frac{b_{R2}\cdot b_{R3}}{|F_{21}|^2} = \frac{11\times 19}{21^2} = \frac{209}{441} \approx 0.4739, \tag{24}$$

where $b_{R2} = b_0 + N_{\mathrm{fam}} - 1 = 11$ (with $b_0 = 7$ the QCD beta-function coefficient and $N_{\mathrm{fam}} = 5$) and $|F_{21}| = N_{\mathrm{gen}} \cdot b_0 = 3 \times 7 = 21$. This corresponds to $+0.23\sigma$ from the NuFIT 6.0 IC24 NH central value, placing it within the 1σ range. Two independent algebraic routes both yield 209/441: Route A, $(b_{R2}/|F_{21}|)\times(b_{R3}/|F_{21}|) = (11/21)\times(19/21)$; Route B, $(b_{R3}/b_{L2}) \times (|F_{21}|+1)/|F_{21}|) = (19/42) \times (22/21)$. The arithmetic identity $2b_{R2} = |F_{21}| + 1$ follows from $N_{\mathrm{gen}} = N_{\mathrm{fam}} - 2$ and is machine-certified (CatAL: `two_b_`

R2_eq_F21_succ, pmns_atm_nlo_candidate; NeutrinoSector.lean, zero sorry, Lean 4). The derivation of (24) from the F_{21} orbit integral at the Lagrangian level is a current research direction (CatB).

The CP-violating phase is derived from $\mathbb{Z}_7$ winding: pmns_cp_phase_from_z7_winding, pmns_cp_phase_degrees: $\delta_{\rm CP} = 205.71°$ (consistent with PDG direction $\delta \approx 197°$).

The Jarlskog invariant follows from the orbit-ratio formulas:

$$J_{\rm GTE} = -\frac{8184\sqrt{437}}{5057221}\sin\left(\frac{\pi}{7}\right) \approx -1.468 \times 10^{-2}, \tag{25}$$

where $5057221 = 13 \times 73^3$, $8184 = 2^3 \times 3 \times 11 \times 31$, $437 = 19 \times 23$. Machine-certified: gte_jarlskog, jarlskog_rational_part, gte_jarlskog_is_negative (NeutrinoSector.lean, zero sorry, Lean 4). Note: $|J_{\rm GTE}|$ reflects GTE's $\delta_{\rm CP}$ prediction; NuFIT 6.0 IC24 NH gives $|J| \approx 1.78 \times 10^{-2}$.

Structural theorems. Additional Lean-certified structure (NeutrinoSector.lean, all zero sorry, Lean 4 [19]):

- fn_dirac_yukawa_rank_theorem: the Dirac Yukawa matrix $Y_D = q_L \otimes q_R$ factors as a rank-1 outer product; any real positive-charge additive FN model saturates rank-1, explaining why standard 3×3 diagonalisation does not apply directly.
- rh_lh_chiral_sectors_distinct: RH and LH sectors are topologically distinct.
- va_forces_distinct_chiral_layers: V−A asymmetry forces layer separation.
- rh_neutrino_couples_antiflavon: RH neutrino couples to antiflavon Φ^*.
- real_yukawa_gives_zero_leptogenesis_cp: real Yukawa matrices give zero leptogenesis CP violation, motivating the kink overlap mechanism for η_B.

9 Conclusion

We have presented a parameter-free structural prediction of the neutrino mass-squared splitting ratio from the Universal Generative Principle's Braid Atlas. The right-handed neutrino b-values $\{5, 11, 19\}$ satisfy $m_{\nu_g} \propto b_g^{29/9}$, giving $\Delta m^2_{21}/\Delta m^2_{31} = 0.02936$, matching NuFIT 6.0 [8] to 0.16σ (0.52%) with no free parameters and with normal ordering automatic.

The exponent 29/9 is over-determined by three independent structural decompositions in terms of the QCD colour rank $N_c = 3$, the charged-lepton mirror offset $\delta = 7$, and the SO(10)/SU(5) GUT representation dimensions — all Lean-certified in ugp-lean [18, 19]. The same integer 29 controls both the seesaw exponent and the structural Dirac Yukawa scale $E_D = v_H/29$, placing $\sum m_\nu \in [40, 100]$ meV within the Planck bound.

The framework makes five preregistered, experimentally falsifiable predictions (F1–F5), testable by JUNO (ratio to 1%), DUNE and Hyper-Kamiokande (ordering), CMB-S4 and Euclid (mass sum), and LEGEND-1000 (Majorana mass). The most important near-term test is the JUNO determination of $\Delta m^2_{21}/\Delta m^2_{31}$: a result outside $[0.028, 0.031]$ would falsify the framework, while confirmation at 0.4% precision would constitute strong support.

Code and Data Availability

All computational verification scripts are available at https://github.com/novaspivack/ugp-physics under `papers/01_SM/canonical_run/`. The relevant scripts are: `comp_p01_EBF_17` (neutrino survey), `EBF_18` (126 bridge), `EBF_19` (29/9 derivation), `EBF_20` (absolute scale), `EBF_21` (structural decomposition), `EBF_22` (full mechanism), `EBF_24` (SO(10) CG / Majorana), and the Froggatt–Nielsen texture identification `comp_p21_SP2_fn_texture_b29_9` (SHA-256 pre-committed prediction block; verdict `TEXTURE_3_2_FROM_Nc`). Each script emits a JSON artifact with a SHA-256 hash; full filenames and descriptions are in `papers/01_SM/PROVENANCE.md` and `papers/21_neutrino_masses/PROVENANCE.md`. Lean formalization: `ugp-lean` [18, 19] (see companion formalization paper [18], zero `sorry`; `lake build` to verify; Lean 4.29.0-rc6, Mathlib 4.29.0-rc6). Neutrino oscillation data: NuFIT 6.0 [8] (Esteban et al., http://www.nu-fit.org/); NuFIT 5.2 [9] (historical comparison).

Acknowledgments

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

References

[1] N. Abgrall et al. The LEGEND-200 and LEGEND-1000 experiments. *J. Phys. Conf. Ser.*, 1468:012136, 2020. arXiv eprint unverified; the arXiv for LEGEND-1000 design report is 2107.11462 (distinct paper). Check INSPIRE for the correct eprint for this proceedings contribution.

[2] N. Abgrall et al. LEGEND-1000 preconceptual design report. 2021.

[3] N. Aghanim et al. Planck 2018 results. VI. Cosmological parameters. *Astron. Astrophys.*, 641:A6, 2020.

[4] M. Aker et al. Direct neutrino-mass measurement based on 259 days of KATRIN data. *Science*, 388:adq9592, 2025. Published April 11, 2025 (arXiv preprint June 2024; key retains 2024 for backward compatibility).

[5] F. An et al. Neutrino physics with JUNO. *J. Phys. G*, 43:030401, 2016.

[6] Sundance O. Bilson-Thompson. A topological model of composite preons. *arXiv preprint*, 2005. arXiv submitted March 22, 2005 (originally listed year 2006 in bib was wrong).

[7] Sundance O. Bilson-Thompson, Fotini Markopoulou, and Lee Smolin. Quantum gravity and the standard model. *Classical and Quantum Gravity*, 24(16):3975–3993, 2007.

[8] I. Esteban, M.C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J.P. Pinheiro, and T. Schwetz. NuFit-6.0: updated global analysis of three-flavor neutrino oscillations. *JHEP*, 12:216, 2024. NuFIT 6.0 global fit (2024), http://www.nu-fit.org/.

[9] I. Esteban, M.C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, and A. Zhou. NuFIT 5.2 (2022). *JHEP*, 09:178, 2020. Updated results available at http://www.nu-fit.org/.

[10] M. Gell-Mann, P. Ramond, and R. Slansky. Complex spinors and unified theories. In P. van Nieuwenhuizen and D. Freedman, editors, *Supergravity*, pages 315–321. North-Holland, 1979.

[11] P. Minkowski. $\mu \to e\gamma$ at a rate of one out of 10^9 muon decays? *Phys. Lett. B*, 67:421–428, 1977.

[12] N. Spivack. Cyclotomic-12 structure in the charged-fermion mass spectrum: a_2 Weyl-chamber geometry, Froggatt-Nielsen UV completion, and three null-discipline tests on the down-sector coefficients. Zenodo, 2026. Paper P19 in the UGP Physics series. https://doi.org/10.5281/zenodo.20168805.

[13] N. Spivack. A deterministic number-theoretic framework for the standard model parameter spectrum. Zenodo, 2026. Paper P01 in the UGP Physics series. https://doi.org/10.5281/zenodo.20168787.

[14] N. Spivack. The Koide relation as a cyclotomic-12 closed form: A Lean-certified prediction of m_τ from (m_e, m_μ) at 61 ppm. Zenodo, 2026. Paper P18 in the UGP Physics series. https://doi.org/10.5281/zenodo.20168795.

[15] Nova Spivack. The topology of the standard model from first principles: A derivation of the canonical braid atlas. Zenodo, 2025. https://doi.org/10.5281/zenodo.20169984.

[16] Nova Spivack. Cosmological predictions of the GTE/Φ_{MDL} framework. Preprint (Zenodo), 2026. https://doi.org/10.5281/zenodo.20465803. UGP Physics series, P47.

[17] Nova Spivack. PSC-optimality of the standard model in 4D gauge theory space. NEMS/PSC Suite, Paper 05, 2026. https://doi.org/10.5281/zenodo.19429721. Two-Layer PSC Theorem: hard constraints force $SU(3) \times SU(2) \times U(1)$; Presentation Invariance selects 3 generations. Machine-checked in nems-lean.

[18] Nova Spivack. ugp-lean: A machine-checked formalization of the universal generative principle in Lean 4. Zenodo, 2026. Formalization paper (latest version): https://doi.org/10.5281/zenodo.19554688. Concept DOI (always-latest): https://doi.org/10.5281/zenodo.19433538. 117 modules across twelve architectural layers including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3-Newton flow, binary cascade, physical mass spectrum), BraidAtlas (ChargeTheorem, CompositeTriples, ChiralitySquaring, ChargeDerivation, CoxeterConductor, CoxeterConductorTowerLaw, EWBosons), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. Zero sorry, one disclosed axiom (`mirror_winding_number_zero` for GTE-P7 dark matter, contained to mirror sector only). Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

[19] Nova Spivack. ugp-lean: Lean 4 formalization of UGP and GTE. `https://github.com/novaspivack/ugp-lean`, 2026. `https://doi.org/10.5281/zenodo.19554700`. 117-module Lean 4 library (module count updated as of 2026-05-09). `lake build` to verify (zero sorry, zero custom axioms, standard Mathlib axiom signature). Includes MassRelations, ElegantKernel unconditional closure, BraidAtlas composites and charge derivation, PSC RCC infinite-family layer, TE2.2 scan certification. New in GTE.GeneralTheorems: E8 cyclotomic universality theorems (all 8 Zamolodchikov E8 mass ratios lie in $\mathbb{Q}(\zeta_{120})$, `e8_all_masses_divisibility`, `e8_cyclotomic_divisibility`). Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

[20] T. Yanagida. Horizontal gauge symmetry and masses of neutrinos. In *Proceedings of the Workshop on the Baryon Number of the Universe and Unified Theories*, pages 95–99, 1979.